

WASP-84b

R-1151

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-84_220122 · created 2026-07-31T01:07:08+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459601.7391 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.013
Transit depth (Rp/Rs)^2	2.67 +/- 0.41 [%]
Orbital Inclination (inc)	88.89 +/- 0.31
Airmass coefficient 1 (a1)	1.0665 +/- 0.0071
Airmass coefficient 2 (a2)	-0.054 +/- 0.031
Scatter in the residuals of the lightcurve fit is	0.67 %
Best Comparison Star	None
Optimal Aperture	4.8
Optimal Annulus	14.91
Transit Duration (day)	0.0535 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.163 $\pm$ 0.013 vs 0.1295 (deviation 2.22 σ)
Duration fit vs published	0.0535 d vs 0.1146 d (deviation 11.2 σ)
Mid-time O-C	-10.7 min
Depth SNR	10.8
Residual scatter	0.67 %

Light curve

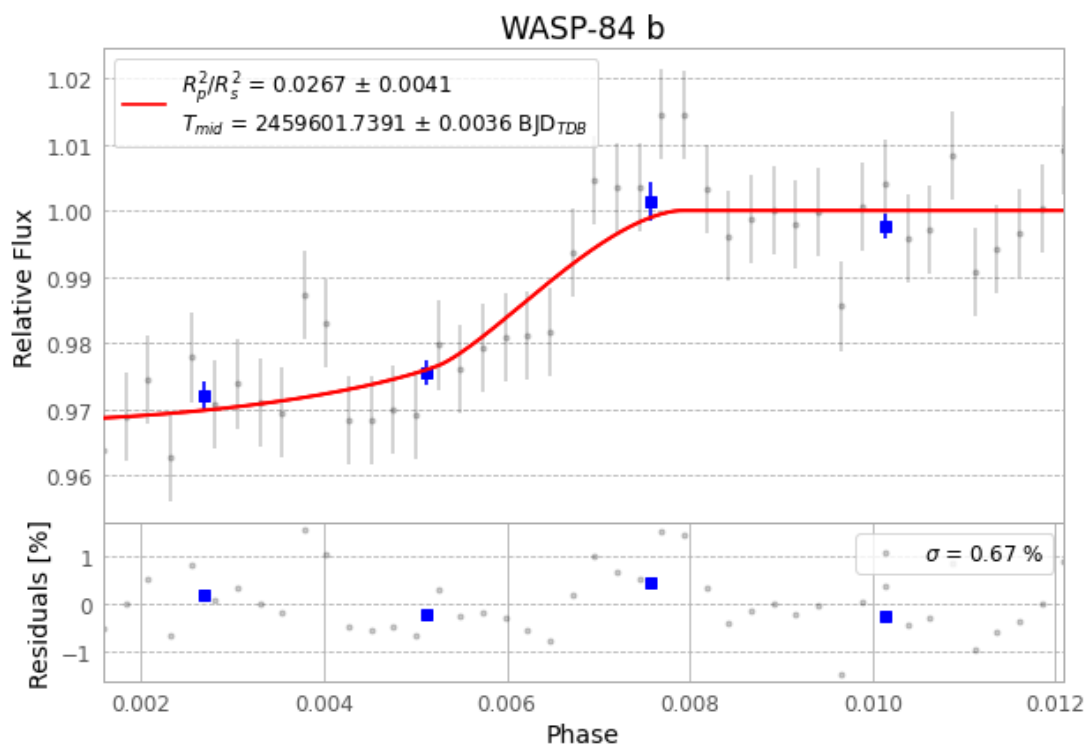


Figure 1 — FinalLightCurve_WASP-84 b_2022-01-21.png (SHA-256 d4f0ce469ce7c7f0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [227, 250], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 20268ccde64cb89f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

44 FITS frames (29 MB), WASP-84220122055412.FITS → WASP-84220122080312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-84-220122.log (d3e3cf8e2576...), FinalLightCurve_WASP-84 b_2022-01-21.png (d4f0ce469ce7...), FinalLightCurve_WASP-84 b_2022-01-21.csv (ae87e30860d2...)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 01:07Z.